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PROBLEMS OF PREMATURITY

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OUTLINE

- Introduction
- Causes/risk factors of preterms and LBW
- Problems associated with preterms and LBW
- General management of premature babies

Newborn classification

1. Based on GA

A. Preterm:

- ✓ Infants born before 37 wk of GA.

B. Term:

- ✓ Infants born b/n 37 & 42 wks of GA.

C. Postterm:

- ✓ Infants born after 42 wks of GA.

Conti...

2. **Based on birth weight (BW)**

A. Normal BW (NBW):-2500 gm – 4000 gm

B. Low BW (LBW):-1500 – 2500 gm

C. Very LBW(VLBW):-1000 – 1500 gm

D. Extremely LBW(ELBW):- <1000 gm

E. Macrosomic:- > 4000 gm

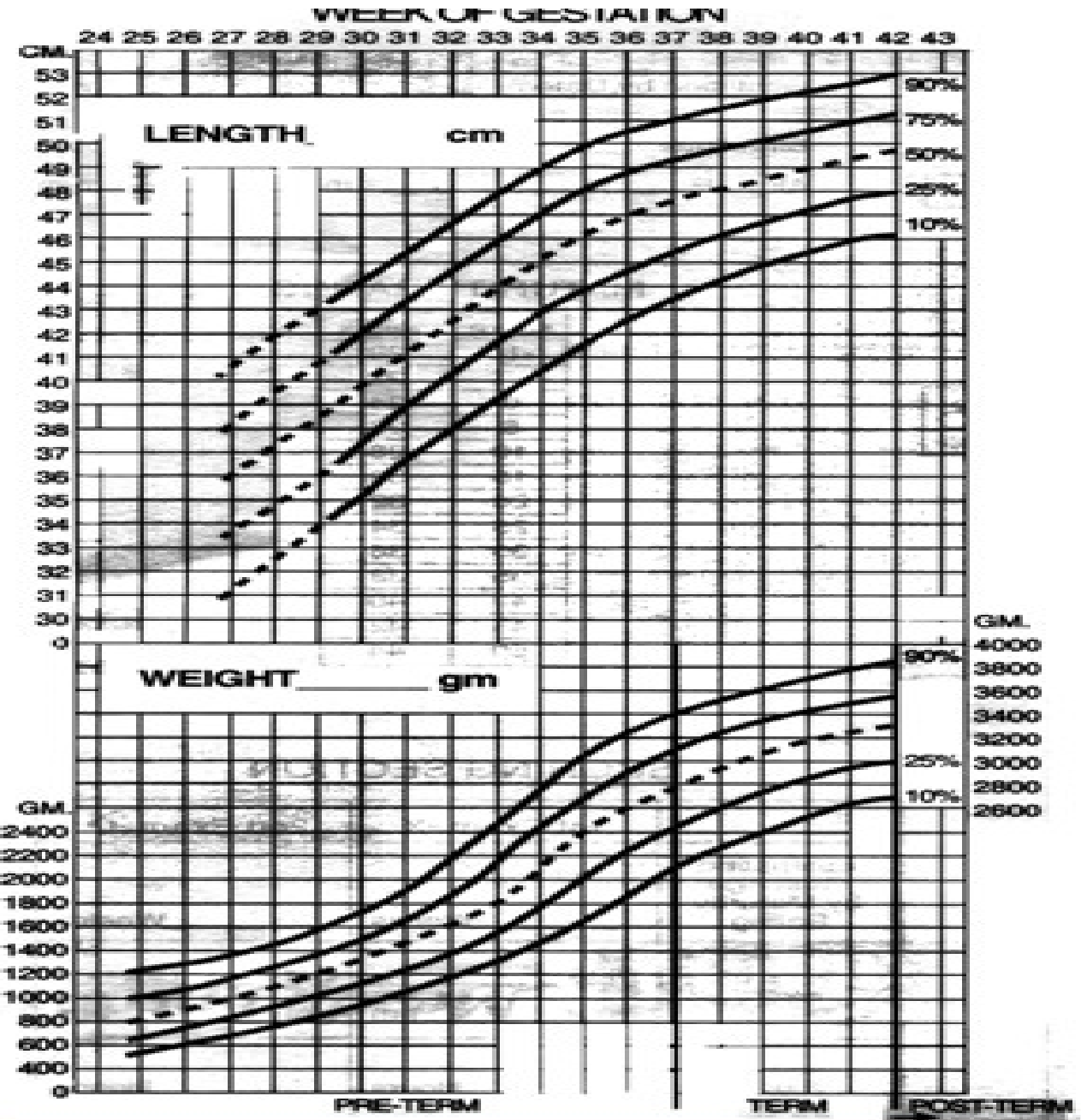
Conti...

3. Based on BW in relation to GA

A. AGA:- b/n 10-90th percentile

B. SGA:- less than 10th centile

C. LGA:- greater than 90th centile



Prematurity and LBW

Definition:

- **Preterms**:- Live born infants delivered before 37 wk from the 1st day of the last menstrual period.
- **LBW**:-is birth weight less than 2500 gram
- **LBW** is due to **prematurity** OR poor intrauterine growth (**IUGR / SGA**), or **both**.
- Prematurity and IUGR are associated with increased neonatal morbidity and mortality.

IDENTIFIABLE CAUSES OF PRETERM BIRTH

- **FETAL**

- ✓ Fetal distress
- ✓ Multiple gestation
- ✓ Erythroblastosis
- ✓ Nonimmune hydrops

- **PLACENTAL**

- ✓ Placental dysfunction
- ✓ Placenta previa
- ✓ Abruptio placenta

Conti...

- UTERINE

- ✓ Bicornuate uterus
- ✓ Incompetent cervix (premature dilatation)

- MATERNAL

- ✓ Preeclampsia
- ✓ Chronic medical illness (cyanotic heart disease, renal disease)
- ✓ Infection (*Listeria mono*, GBS, UTI, bacterial vaginosis, chorioamnionitis)
- ✓ Drug abuse (cocaine)

Conti...

- OTHERS

- ✓ Premature rupture of membranes

- ✓ Polyhydramnios

- ✓ Iatrogenic

- ✓ Trauma

NEONATAL PROBLEMS ASSOCIATED WITH PREMATURE INFANTS

➤ METABOLIC-ENDOCRINE

- Hypoglycemia
- Hypothermia
- Hypocalcemia
- Hyperglycemia
- Late metabolic acidosis
- Euthyroid but low-thyroxin status

➤ INFECTIONS

- Due to reduced immunity

Conti...

➤ RESPIRATORY

- Respiratory distress syndrome (hyaline membrane disease)
- Apnea
- Bronchopulmonary dysplasia
- Pneumothorax,
- pneumomediastinum
- interstitial emphysema
- Congenital pneumonia
- Pulmonary hypoplasia
- Pulmonary hemorrhage

Conti...

➤ CARDIOVASCULAR

- Patent ductus arteriosus
- Hypotension
- Hypertension
- Bradycardia (with apnea)
- Congenital malformations

Conti...

➤ HEMATOLOGIC

- Hyperbilirubinemia-indirect
- Subcutaneous, organ (liver, adrenal) hemorrhage
- Disseminated intravascular coagulopathy
- Vitamin K deficiency
- Anemia of prematurity (early or late onset)

Conti...

➤ GASTROINTESTINAL

- Neonatal Necrotizing enterocolitis
- Poor gastrointestinal function-poor motility
- Hyperbilirubinemia-direct and indirect
- Congenital anomalies producing polyhydramnios
- Spontaneous gastrointestinal isolated perforation

Conti...

➤ RENAL

- Hyponatremia
- Hypernatremia
- Hyperkalemia
- Renal tubular acidosis
- Renal glycosuria
- Edema

Acute Complication of prematurity

- Respiratory distress syndrome (HMD)
- Hypoglycemia
- Necrotizing enterocolitis
- Apnea of prematurity
- Hypothermia
- Sepsis
- Intraventricular hemorrhage
- Jaundice

late onset complications of prematurity

- Anemia of prematurity
- Retinopathy of prematurity
- Osteopenia of prematurity
- Bronchopulmonary dysplasia
- Developmental delay
- Mental retardation
- Hearing and vision deficit
- Patent ductus arteriosus

1. Respiratory Distress Syndrome

- Respiratory distress syndrome (RDS), formerly known as hyaline membrane disease (HMD),
- Describes a disease typical of preterm infants that is caused by insufficient pulmonary surfactant in alveoli.

Etiology

➤ Surfactant deficiency

- ✓ Decreased production and secretion- is the primary cause of RDS.

Incidence

- Respiratory distress syndrome (RDS) occurs primarily in premature infants.
- Its incidence is inversely related to gestational age and birth weight.
- It occurs in:-
 - ✓ 60- 80% of infants <28 wk GA
 - ✓ 15–30% of those between 32 and 36 wk GA and
 - ✓ Rarely in those >37 wk GA.

Risk factors

➤ The risk for development of RDS increases with:

A. Fetal sex

✓ Male infants are at higher risk for RDS

- Due to the presence of **circulating weak fetal androgens**
- **Inhibit the production** of surfactant phospholipids.

B. Maternal diabetes

✓ Due to **enhanced production of fetal insulin**

- **Inhibits the production of proteins** important for surfactant function.

Conti...

C. Multiple births

D. Cesarean delivery

E. Precipitous delivery

F. Asphyxia

G. Cold stress

H. Maternal history of previously affected infants

I. Mutations in surfactant-related proteins

Conti...

➤ The risk of RDS is reduced in:-

- ✓ Pregnancies with chronic or pregnancy-associated hypertension
- ✓ Maternal heroin use
- ✓ Prolonged rupture of membranes
- ✓ Antenatal corticosteroid prophylaxis.

Pathophysiology

- As a consequence of progressive injury to epithelial and endothelial cells from atelectasis (atelectrauma), volutrauma, ischemic injury, and oxygen toxicity, effusion of proteinaceous material and cellular debris into the alveolar spaces (forming the classic hyaline membranes) further impairs oxygenation.
- Alveolar atelectasis, hyaline membrane formation, and interstitial edema make the lungs less compliant in RDS, so greater pressure is required to expand the alveoli and small airways.
- Although surfactant is present in high concentrations in fetal lung homogenates by 20 wk of gestation, it does not reach the surface of the lungs until later.
 - ✓ It appears in amniotic fluid between 28 and 32 wk.
 - ✓ Mature levels of pulmonary surfactant are present usually after 35 wks. of gestation.

Conti...

- Atelectasis results in perfused but not ventilated alveoli, causing hypoxia.
- Decreased lung compliance, small tidal volumes, increased physiologic dead space, and insufficient alveolar ventilation eventually result in hypercapnia.
- The combination of hypercapnia, hypoxia, and acidosis produces pulmonary arterial vasoconstriction with increased right-to-left shunting through the foramen ovale and ductus arteriosus and within the lung itself.

Conti...

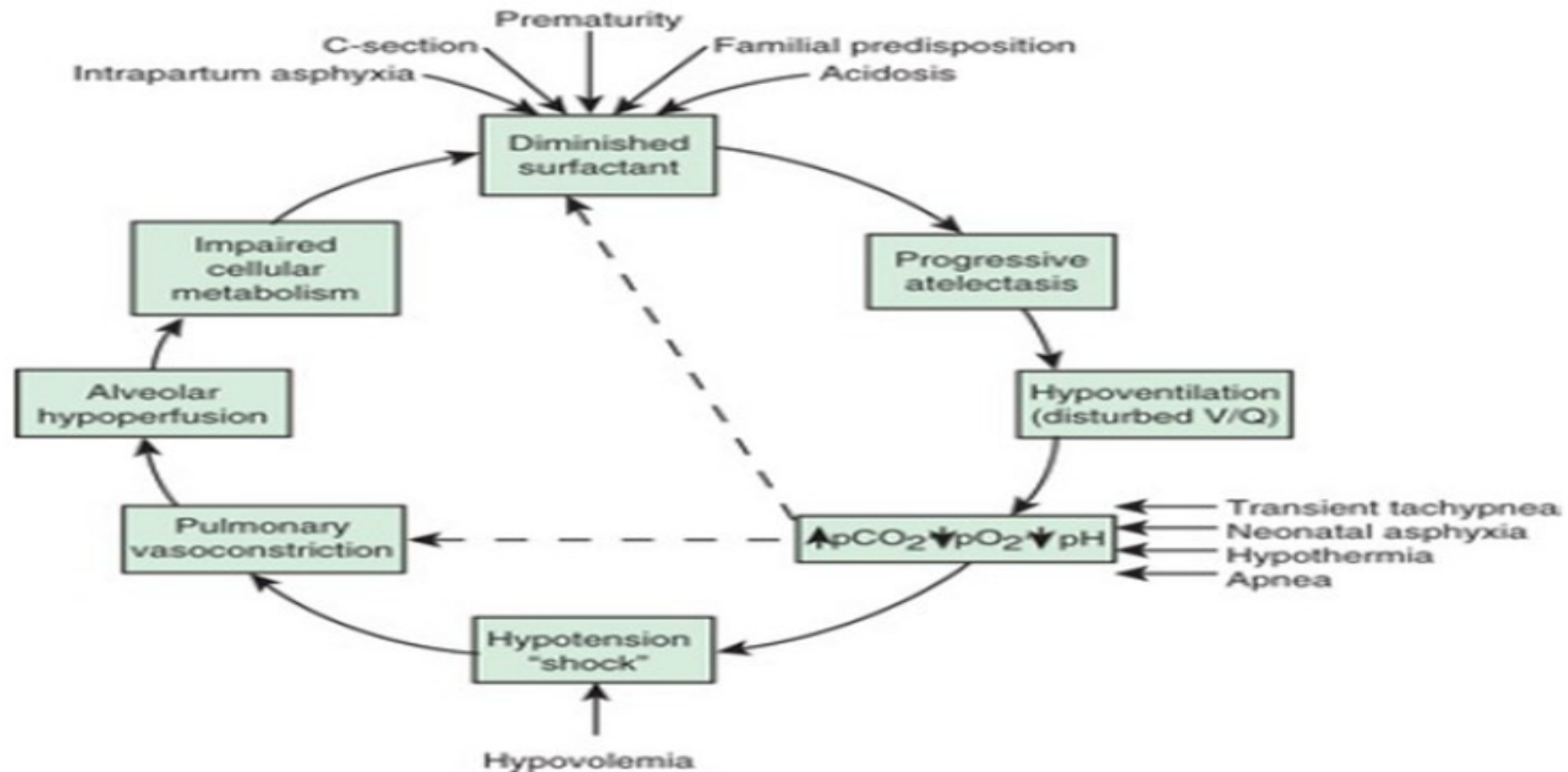


FIG. 122.4 Contributing factors in the pathogenesis of hyaline membrane disease. The potential "vicious circle" perpetuates hypoxia and pulmonary insufficiency. V/Q, Ventilation-perfusion ratio. (From Farrell P, Zachman R: Pulmonary surfactant and the respiratory distress syndrome. In Quilligan EJ, Kretchmer N, editors: *Fetal and maternal medicine*, New York, 1980, Wiley. Reprinted by permission of John Wiley and Sons, Inc.)

Conti..

- The **major constituents of surfactant** are dipalmitoyl phosphatidylcholine (lecithin), phosphatidylglycerol, apoproteins (surfactant proteins SP-A, SP-B, SP-C, and SP-D), and cholesterol.
- ✓ Surfactant recovered from lungs of all mammalian species by alveolar wash procedures contains **70% to 80% phospholipids, about 8% to 10% protein**, and about **10% neutral lipids, primarily cholesterol**.
- ✓ The **phosphatidylcholine** species of the phospholipids represents about 60% by weight of the surfactant and about 80% of the phospholipids.
- ✓ It is synthesized, packaged, and secreted from alveolar type II cells of the lung.

Conti...

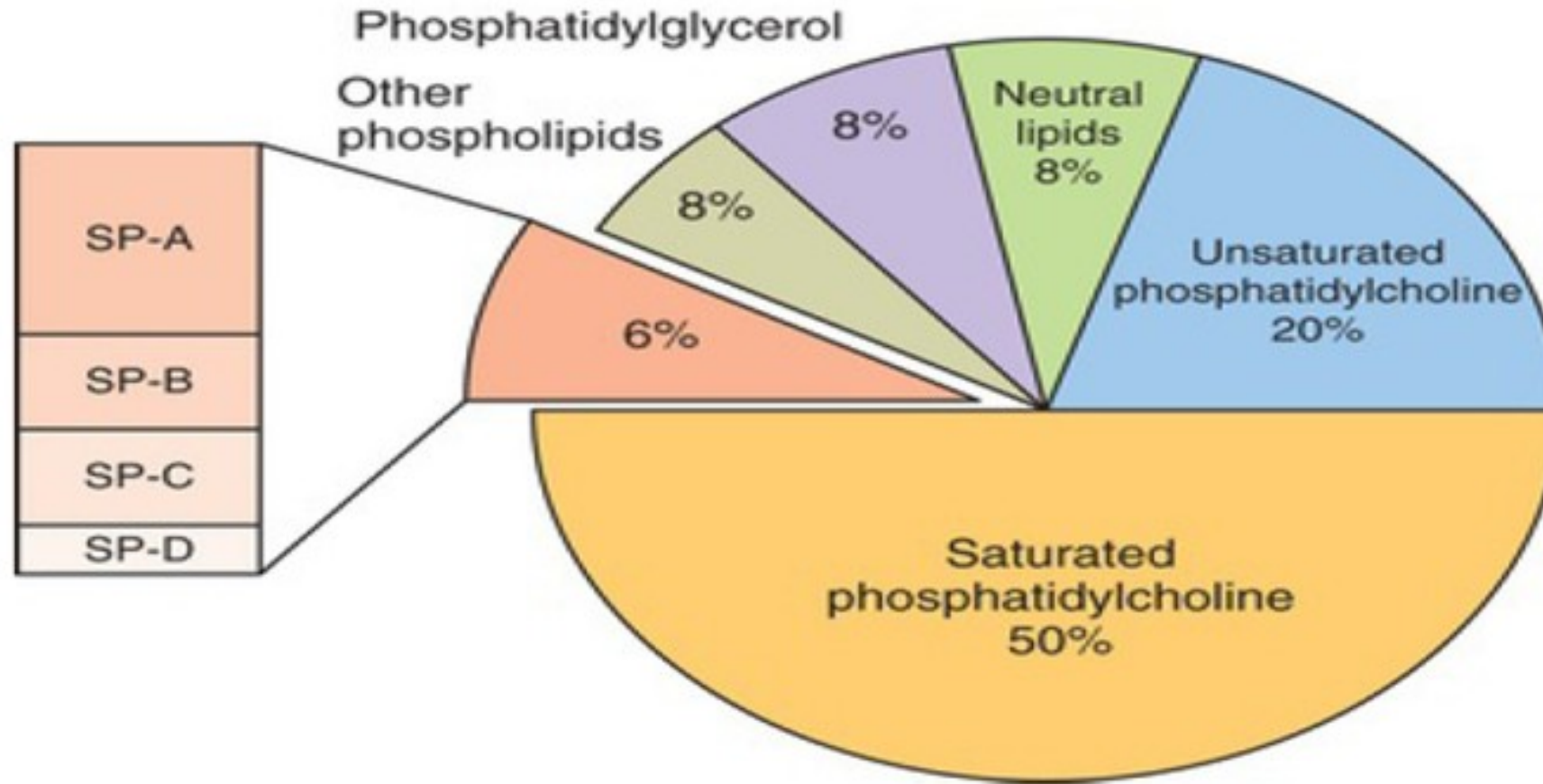


FIG. 122.2 Composition of surfactant. SP-A, Surfactant-associated protein A; SP-B, surfactant-associated protein B; SP-C, surfactant-associated protein C; SP-D, surfactant-associated protein D. (From Jobe AH, Ikegami M: Biology of surfactant, *Clin Perinatol* 28:655–669, 2001.)

Clinical Manifestations

- Signs of RDS usually appear **within minutes of birth**, although they may not be recognized for several hours in larger premature infants, until rapid, shallow respirations become more obvious.
- A later onset of tachypnea should suggest other conditions.
- Some patients require resuscitation at birth because of intrapartum asphyxia or initial severe respiratory distress (especially with birthweight <1,000 g).

Conti...

- **Characteristically, tachypnea, prominent (often audible) expiratory grunting, intercostal and subcostal retractions, nasal flaring, and cyanosis are noted.**
- Breath sounds may be **normal or diminished with a harsh tubular quality**, and on deep inspiration, fine crackles may be heard.
- The natural course of untreated RDS is characterized by progressive worsening of cyanosis and dyspnea.

Conti...

- If the condition is inadequately treated, blood pressure may fall; cyanosis and pallor increase, and grunting decreases or disappears, as the condition worsens.
- ***Apnea and irregular respirations*** are ominous signs requiring immediate intervention.
- Untreated patients may also have a mixed respiratory-metabolic acidosis, edema, ileus, and oliguria.
- Respiratory failure may occur in infants with rapid progression of the disease.

Conti...

- In most cases the signs reach a **peak within 3 days**, after which improvement is gradual.
- Improvement is often heralded by **spontaneous diuresis** and improved blood gas values at lower inspired O₂ levels and/or lower ventilator support.
- Death can result from severe impairment of gas exchange, alveolar air leaks (pulmonary interstitial emphysema, pneumothorax), pulmonary hemorrhage, or intraventricular hemorrhage (IVH).

Downe's score for grading severity of respiratory distress

Downes' score	0	1	2
Respiratory rate	<60Breaths per min	60-80	>80
Retractions	None	Mild	Severe
Cyanosis	None	Cyanosis relieved by oxygen	Cyanosis not relieved by oxygen
Grunting	None	Audible with stethoscope	Audible with ear
Air entry	Good bilaterally	Mildly decreased	Markedly decreased

Grades of severity according to the total score: 0 – 3 - mild; 4 –6 - moderate, >6 severe

Natural course

- **The natural course of untreated RDS** is characterized by:
 - ✓ Progressive worsening of cyanosis and dyspnea.
 - ✓ If the condition is inadequately treated, blood pressure may fall; cyanosis and pallor increase, and grunting decreases or disappears, as the condition worsens.
 - ✓ Apnea and irregular respirations are ominous signs requiring immediate intervention.
 - ✓ Untreated patients may also have a mixed respiratory-metabolic acidosis edema, ileus, and oliguria. Respiratory failure may occur in infants with rapid progression of the disease.
 - ✓ In most cases the signs reach a peak within 3 days, after which improvement is gradual. Improvement is often heralded by spontaneous diuresis and improved blood gas values at lower inspired O₂ levels and/or lower ventilator support.

Diagnosis

- The clinical course, chest x-ray findings, and blood gas values help establish the clinical diagnosis.
- **On chest radiograph, the lungs may have a characteristic but not pathognomonic appearance that includes low lung volumes, a diffuse, fine reticular granularity of the parenchyma (ground-glass appearance), and air bronchograms.**
- The initial x-ray appearance is occasionally normal, with the typical pattern developing during the 1st day.
- Considerable variation in radiographic findings may be seen, especially in infants who have already received treatment with surfactant replacement and/or positive pressure respiratory support; this variation often results in poor correlation between radiographic findings and the clinical course.

Conti...

- Blood gas findings are characterized initially by hypoxemia and later by progressive hypoxemia, hypercapnia, and variable metabolic acidosis.
- In the differential diagnosis, early-onset sepsis may be indistinguishable from RDS.
- In neonates with pneumonia, the chest radiograph may be identical to that for RDS.
- Clinical factors such as maternal group B streptococcal colonization with inadequate intrapartum antibiotic prophylaxis, maternal fever ($>38.6^{\circ}\text{C}$) or chorioamnionitis, or prolonged rupture of membranes (>12 hr) are associated with an increased risk of early-onset sepsis.
- Although complete blood counts are neither sensitive nor specific in the diagnosis of early-onset sepsis, the presence of marked neutropenia has been associated with increased risk.

Chest x-ray finding in RDS

➤ Chest X-ray – have 4 grades

- Grade 1- Fine homogeneous reticulogranular pattern (Homogeneous micro atelectasis).
- Grade 2- Widespread air bronchograms become visible.
- Grade 3- Cardio pulmonary differentiation difficult due to alveolar shadowing.
- Grade 4- Complete white-out of the lung fields.

Diagnosis

- Shake test
- Arterial blood gas analysis
- Chest X ray
- Septic work up
- Echocardiography

Prevention

- Avoidance of unnecessary or poorly timed **early cesarean delivery** or induction of labor, appropriate management of high-risk pregnancy and labor.
(including administration of antenatal corticosteroids), and prediction of pulmonary immaturity with possible in utero acceleration of maturation are important preventive strategies.

Conti...

- Administration of antenatal corticosteroids to women before 37 wk gestation significantly reduces the incidence and mortality of RDS as well as overall neonatal mortality.
- Antenatal steroids also reduce:- Admission to the neonatal intensive care unit (NICU) and need for/duration of ventilatory support
 - ✓ Incidence of severe IVH
 - ✓ Necrotizing enterocolitis (NEC)
 - ✓ Neurodevelopmental impairment.
 - ✓ Overall mortality.
- For all women between 24 and 36 wk gestation who present in preterm labor and are likely to deliver a fetus within 1 wk, antenatal corticosteroid administration should be considered.

Mechanism of action

- Corticosteroid stimulation of **developmentally regulated gene expression** and **physiologic functions result in maturation** of the lungs and some other tissues.
- Antenatal steroids **accelerate development of type 1 and type 2 pneumocytes**.
 - ✓ leading to **structural and biochemical changes** that improve **both lung mechanics** (maximal lung volume, compliance) and **gas exchange**.
- **Induction of type 2 pneumocytes increases surfactant production**
 - ✓ By **inducing production of surfactant proteins and enzymes** necessary for phospholipid synthesis.

Conti...

- Other effects of antenatal corticosteroids include:-
 - ✓ Induction of pulmonary beta-receptors, which play a role in **surfactant release** and **absorption of alveolar fluid** when stimulated
 - ✓ Induction of fetal lung antioxidant enzymes
 - ✓ Upregulation of gene expression for the epithelial Na⁺ channel, which is important for **absorption of lung fluid after birth.**
- For these changes to occur, however, the lungs need to have **reached a stage of development that is biologically responsive to corticosteroids.**

Steroid

- Indicated for 24-34 weeks gestation age, (22+0 to 33+6 weeks), (< 36 plus 6 WEEKS)
- Betamethasone or Dexamethasone.
- Dose:
 - ✓ 2 doses of betamethasone are administered 12mg IM at an interval of 24 hours or
 - ✓ 4 doses of dexamethasone, 6mg each at an interval of 12 hours
- Effect starts 24 hours (possibly as early as 4 hours) after 1st dose and continues for 7 days.
- A repeat course of steroids is to be considered if it was taken > 7 days ago and gestation is < 32 weeks and delivery is likely.

Management principle

- Basic supportive care & Monitoring
- NCPAP
- Mechanical ventilators
- Administer surfactant
- Address fluid and vasopressors
- Antibiotics
- Feeding

2. Hypoglycemia

INTRODUCTION

- Hypoglycaemia is one of the most common metabolic problems seen in NICU.
- The **definition of hypoglycaemia** as well as its clinical significance and management remain **controversial**.
- Blood glucose levels in the first hours of life are typically lower than normal values of older children or adults.

Conti...

- During the normal transition to extrauterine life, blood glucose concentration in the healthy term newborn **falls during the first two hours after delivery.**
 - ✓ **Reaching a nadir (first 2 hrs)** that usually is no lower than 40 mg/dL.
- It is important to differentiate this normal physiologic transitional response from disorders that result in persistent or recurrent hypoglycemia, which if left untreated may lead to significant neurologic and developmental sequelae.

Conti...

- The thresholds for treating hypoglycaemia depend on:-
 - ✓ The presence of symptoms
 - ✓ The age of the infant in hours
 - ✓ The persistence of hypoglycaemia.
- According to the Paediatric Endocrine Society, by **48 to 72 hours** of life glucose control should be **similar to that of older children and adults** (should be **>60 mg/dL**).
- Bedside **whole blood glucose** measurements are 15% lower than **plasma levels**.

Common causes of hypoglycaemia in neonates

➤ ↓ Glucose stores & ↓ production

- ✓ Preterm

- ✓ IUGR or SGA

- ✓ ↓ Caloric intake

- ✓ Delayed feeding

➤ ↑ Glucose utilization (hyperinsulinism)

- ✓ IDMs or LGA infants

- ✓ Erythroblastosis fetalis

- ✓ Abrupt cessation of high glucose intake

- ✓ Beckwith-Wiedemann syndrome

- ✓ Islet cell hyperplasia

- ✓ Insulin producing tumours

- ✓ After exchange transfusion

- ✓ Maternal drugs (intrapartum glucose infusion)

Conti...

➤ **↑Glucose utilization and/or ↓Production**

- ✓ Perinatal stress (hypothermia, sepsis, asphyxia, respiratory distress, shock)
- ✓ Polycythaemia
- ✓ Maternal drugs (β - blockers, steroids)
- ✓ Endocrine deficiency (adrenal haemorrhage, CAH, hypothyroidism)
- ✓ Inborn errors of metabolism (galactosemia, glycogen storage disease, tyrosinemia)
- ✓ Congenital heart diseases

Clinical manifestations

- Signs are **non-specific** and can be similar to signs of many other problems.
- Asymptomatic hypoglycaemia is a common transient problem in most neonates.
- A smaller proportion of infants with hypoglycaemia can be symptomatic.
- **Symptomatic hypoglycaemia** is an emergency and requires IV treatment.
- **Symptoms** include: jitteriness, tremors, irritability, seizures, coma, apnoea, cyanosis, lethargy and poor feeding, weak or high-pitched cry, hypothermia and respiratory distress.
- Episodes of sweating, sudden pallor, hypothermia and cardiac arrest have also been reported.

Prevention

- At birth
 - ✓ **Dry the baby**
 - ✓ **Avoid hypothermia**
 - ✓ **Encourage skin-to-skin care.**
- Encourage early enteral feeding (within one hour of age) and frequently thereafter (at least 8 feeds/day).
- Perform **serial RBS monitoring in infants at risk of hypoglycaemia**, starting from the first 1-2 hours of life and do not allow them to wait for >3 hours between feedings.

Management Principle

- Serum glucose levels should always be evaluated and treated.
- Management includes **anticipation of neonates at risk, correction and investigation and treatment of the cause.**
- **Asymptomatic hypoglycaemia: 25mg/dl to 45mg/dl**
 - ✓ Enterally feeding
 - ✓ Start enteral feeding at 60-80mL/kg/day if there are no contra-indications.
 - ✓ If persistent, consider admission to NICU and administration of IV fluid
- **Symptomatic and if RBS is < 25 mg/dl = IV Dextrose 10% , and then after according to follow management protocol.**

3. Apnea of prematurity

Definition:

➤ Apnea is defined as:-

- ✓ The **cessation of airflow or cessation of breathing** for **> 20 sec** or
- ✓ Any duration associated with cyanosis/bradycardia <100/min /poor tone.

Incidence

- Incidence of apnea **increases with decreasing gestational age.**
- Essentially, all infants <28 weeks' gestational age have apnea.
- As many as 25% of all premature infants whose weight <1,800 g.
(~34 weeks' gestational age) have at least one apneic episode.
- Apnoea of prematurity generally occurs **one or two days after birth.**
- If it does **not occur during the first seven days, it is unlikely** to occur later.

Causes of apnoea

1. **Idiopathic:** Apnoea of prematurity.
2. **Temperature:** Hypothermia / Hyperthermia.
3. **CNS:** Intracranial haemorrhage, seizures; drugs: depressant, maternal narcotics and magnesium sulphate.
4. **Respiratory:** hypoxia, RDS, pneumonia, obstructive airway lesions, laryngeal reflex, pneumothorax, nasal occlusion and tracheal occlusion caused by neck flexion.
5. **CVS:** heart failure, hypotension, hypertension, hypovolaemia, PDA.
6. **GIT:** GERD, NEC
7. **Infection:** pneumonia, sepsis, meningitis.
8. **Metabolic:** acidosis, hypoglycaemia, hypocalcaemia, hyponatraemia, hypernatremia, inborn errors of metabolism .
9. **Haematological:** anaemia, polycythaemia.

Potential Causes of Neonatal Apnea

Idiopathic	Apnea of prematurity
Central nervous system	Intraventricular hemorrhage, seizures, hypoxic injury, herniation, drugs – depressant, maternal narcotics and magnesium sulphate,
Respiratory	Pneumonia, obstructive airway lesions, upper airway collapse, atelectasis, extreme prematurity, laryngeal reflex, phrenic nerve paralysis, pneumothorax, hypoxia, RDS, nasal occlusion caused by phototherapy eye patches, tracheal occlusion caused by neck flexion
Infectious	Sepsis, meningitis, pertussis
Gastrointestinal	Necrotizing enterocolitis, GERD,
Metabolic	↓ Glucose, ↓ calcium, ↓/↑ sodium, ↑ ammonia, ↑ organic acids, ↑ ambient temperature, hypothermia
Cardiovascular	Hypotension, hypertension, heart failure, anemia, hypovolemia, PDA
Haematological	Anemia, polycythemia
Other	Immaturity of respiratory center, sleep state

Conti...

- **Idiopathic apnea of prematurity** occurs in the absence of identifiable predisposing diseases.
- The incidence of idiopathic apnea of **prematurity varies inversely with gestational age**.
- The onset of idiopathic apnea can be during the **1st 1-2 wk after birth** but is often delayed if there is RDS or other causes of respiratory distress.
- Although **apnea of prematurity** is the most common cause of apnea in preterm infants, it is a **diagnosis of exclusion**.

Classification

➤ **Classification** of apnea is based on whether absent airflow is accompanied by continued inspiratory efforts and upper airway obstruction.

➤ Most spells in preterm infants are mixed.

➤ **There are 3 types.**

1. Central apnea

- ✓ Occurs when **inspiratory efforts are absent**.
- ✓ It is caused by **decreased CNS stimuli to respiratory muscles**.
- ✓ Both airflow and chest wall motion are absent (no airflow, no respiratory efforts).

Conti...

2. Obstructive apnea

- ✓ Occurs when **inspiratory efforts persist** in the **presence of airway obstruction**, usually at the pharyngeal level.
- ✓ It is characterized by **absence of airflow but persistent chest wall motion** (no airflow, despite respiratory efforts).

3. Mixed apnea

- ✓ Occurs (50-75% of cases)
 - ✓ When airway obstruction with inspiratory efforts precedes or follows central apnea.
- Short episodes of apnea are usually central, where as prolonged ones are often mixed.

Monitoring

- **All babies < 34 weeks' of gestational** age should be monitored continuously for at least the first week of life or until there is an absence of apnoeic episodes for at least 7 days.
- **Babies >34 weeks' gestational** age should be monitored if **they are sick**.
- **Pulse oximeter** is an effective way of monitoring apnoea.
 - ✓ Though it does not detect chest wall movement, it detects the most important clinical consequences of apnoea, i.e., **hypoxia and bradycardia**.
 - ✓ When a **monitor alarm sounds**, one should remember to respond to the infant, not to the monitor, **checking for bradycardia, cyanosis, and airway obstruction**.

Management of apnoea

Immediate measures:

- Most apnoeic spells in preterm infants respond to **tactile stimulation**.
- If the infant fails to respond with stimulation, then the infant should be ventilated with a bag and mask.

GENERAL MEASURES

- General measures are **usually preventive in nature and are applied to all infants less than 35 weeks gestation who are at risk for apnea.**
- These interventions are directed towards eliminating factors that increase the risk of apnea or reduce the prevalence of associated hypoxia.

Conti...

- Environmental and temperature control
- **Consider underlying causes of new onset of apnea or increased severity**
 - ✓ Such as neonatal sepsis.
 - ✓ If an underlying factor has been identified, correct it/treat it.
- Head and neck position
 - ✓ Positions of extreme flexion or extension of the neck should be avoided to reduce the likelihood of airway obstruction.
- Maintain nasal patency
 - ✓ Nasal patency is preserved by **avoiding vigorous nasal suctioning** or prolonged use of NG tubes.
 - ✓ Care should be taken to avoid reflexes that may trigger apnoea.
- Oxygen supplementation to maintain oxygen saturation (SpO₂) at 90 to 95 percent
 - ✓ To avoid baseline hypoxemia, which predisposes to episodes of severe desaturation.

METHYLXANTHINE THERAPY

- Methylxanthines cause **stimulation of respiratory neural output**, presumably by inhibiting adenosine receptors
- Are the primary pharmacologic therapy used to treat apnea of prematurity.
- The **two methylxanthines** used in apnea of prematurity are [caffeine](#) and [theophylline](#).
- **Caffeine is the preferred agent** because of its **longer half-life, wider margin of safety, and lower frequency of adverse effects**

4. Necrotizing Enterocolitis (NEC)

Definition.

- NEC is an acute inflammatory injury of the **intestine**.
- The colon appears to be the most commonly affected site.
- Characterized by various degrees of **mucosal or transmural necrosis of the intestine**.
- It is associated with **inflammation, invasion of enteric gas-forming organisms**, and dissection of gas into the muscularis and portal venous system.

Conti...

- The **triad** of **intestinal ischemia** (injury), **enteral nutrition** (metabolic substrate), and **bacterial translocation** has classically been linked to NEC.
- NEC is an acute inflammatory disease in the newborn's intestine characterized by **hemorrhagic necrosis**, which may lead to **perforation** and **destruction of the gut**.

- NEC is the most common surgical emergency in the neonatal period, affecting **0.5%** of all live births and **3–5%** of low-birth-weight live births.
- Although primarily a disease of the preterm infant, 6–13% are in full-term infants.
- NEC incidence in developed countries is 7-11% in infants < 1.5 kg.
- > 85% of cases of NEC occur among preterm infants (< 32 ks' gestation).
- There is a male preponderance of 2:1.

Etiology

- Whilst the precise mechanism of NEC remains unclear, a number of predisposing risk factors have been identified:-
 - ✓ Low birth weight
 - ✓ Prematurity
 - ✓ Enteral feeds
 - ✓ Sepsis.

Risk factors

- **Prematurity, especially < 32 weeks, or VLBW**
- IUGR
- Sepsis
- Hypoxia
- Antenatally detected umbilical artery blood flow abnormalities- (Absent/reversed end– diastolic flow)
- Altered intestinal microbe (dysbiosis)/ prolonged antibiotic use
- Formula feeding
- Lack of antenatal corticosteroid (GA of 24- 34 weeks)
- Use of H2 blockers

Conti...

Prematurity

- Approximately 90% of NEC cases are found in premature infants, with mean gestational age of 30–32 weeks.
- The increased incidence of NEC in this group may be due to:-
 - ✓ Greater susceptibility of the premature gut to the stresses of hypoxia, ischemia and the mucosal damage of hyperosmolar feeds in this group.

EPIDEMIOLOGY AND PATHOGENESIS

- The timing of the onset of NEC varies inversely with gestational age.
- In the severely premature neonate, onset often occurs in the fourth week after birth.
- In neonates born closer to **term**, onset typically occurs in the **first week** of birth.
- The lower the birthweight of a neonate, the greater is the risk of developing NEC.
- Only approximately 10% of cases occur in full-term newborns.

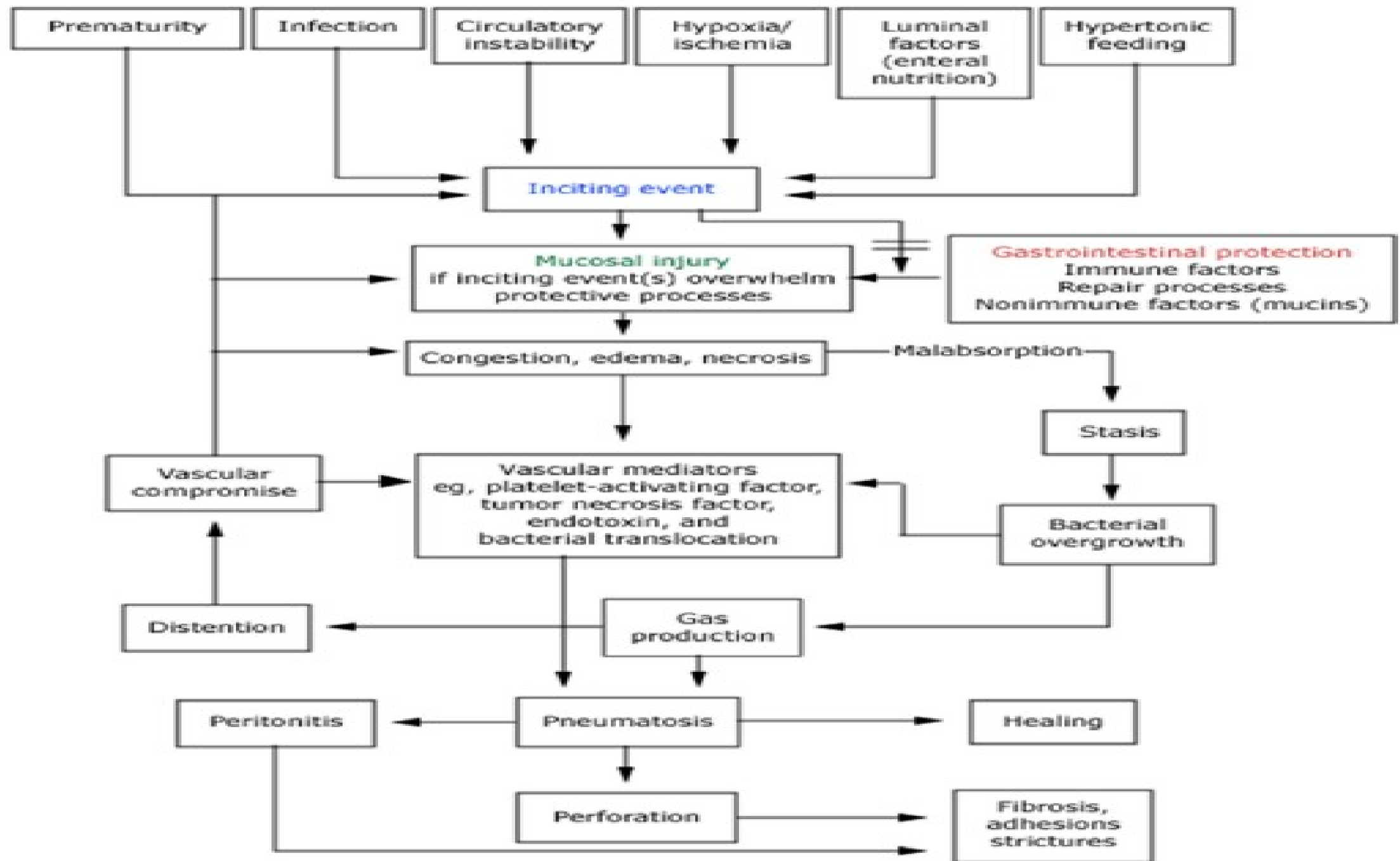
- The precise pathophysiology of NEC remains unclear, but it is considered to be multifactorial in origin.
 - ✓ The culmination of microbial colonization and inflammatory insult with mucosal injury to an intestine that is compromised by hypoxic ischemia and/or immaturity.
- Hypoxia and circulatory ischemia result in preferential distribution of the limited blood supply to vital organs such as the heart, brain and kidney, at the expense of compromising blood flow away from non-vital regions including the splanchnic circulation of the intestinal tract (the Herring–Breur reflex).

Conti...

- Intestinal ischemia and bacterial colonization stimulate proinflammatory mediators (including interferon, tumor necrosis factor (TNF)- α and platelet activating factor (PAF), which through multiple cascades may lead to variable degrees of coagulative and ischemic necrosis of the bowel.
- Furthermore free radicals released from inflammatory cascades following ischemia and reperfusion contribute to the tissue necrosis by causing lipid peroxidative damage to cellular membranes.
- Diffusion of free radicals and the effect of gasforming organisms within the gut wall result in the pathognomonic pneumatosis intestinalis that is seen on radiographs in 90% of cases.
- Transmural necrosis with bacterial translocation across the gut wall may result in perforation with peritonitis, bacteremia or septicemia

Conti...

- Up to **35% of patients with NEC have positive cultures**, with the most common implicated pathogens being **Klebsiella, Escherichia coli, Clostridium difficile and Streptococcus faecalis**.
- The increased propensity of premature infants to develop NEC injury is attributed to a number of additional factors.
- These include a deficient intestinal microbial host defense, poor development of the mucosal barrier, an immature mucosa at greater susceptibility to hyperosmolar feeds and an accentuated pro-inflammatory response.
- If the intestinal injury is short-lived, the pathological course may be followed by healing with fibrosis, with or without a resulting stricture in the affected bowel.
- The disease most commonly involves the terminal ileum and ascending colon, but it may affect variable segments of the small and large intestine, being isolated (30%), multifocal (55%) or pan-intestinal (15%).



Clinical signs and symptoms

- The clinical course of NEC is variable.
- The severity of the disease may be categorized clinically using the original **Bell's staging system for NEC**, a classification later modified by Walsh and Kliegman.

Conti...

Stage I

- Suspected NEC.
- At the early stage, symptoms and signs are usually **non-specific**.
e.g. irritability, fever, apneas and bradycardias, and poor toleration of feeds.
- Many of these children are more likely to be suffering from **feeding intolerance** commonly associated with prematurity or low birth weight.

Conti...

Stage II

- Established NEC.
- These babies will have established **clinical and radiographic signs of NEC** with moderate–profound systemic illness, including **metabolic acidosis and thrombocytopenia**.
- Vomiting/aspirates are usually bilious or bloodstained whilst stools are likely to be blood-stained.
- Abdominal distension, abdominal wall erythema and edema are usual.
- There may or may not be evidence of perforation clinically and/or radiologically.
- A clinically palpable intra-abdominal mass (reported in 11%) may indicate the presence of persistently dilated bowel loops or, if the disease is localized, an intraperitoneal abscess.

Conti...

Stage III

- Advanced NEC with **systemic instability**.
- As well as features of stage II disease, these babies have cardiovascular instability (hypotension), severe metabolic acidosis and coagulative derangements.
- There will usually be abdominal and **radiological evidence of perforation**, peritonitis and ascites with generalized sepsis.

Complications

1. Metabolic disorders including metabolic acidosis and disseminated intravascular coagulation.
2. Perforation with peritonitis and septicemia
3. Localized intraperitoneal abscess formation
4. Healing by fibrosis with stricture formation
(20% in term infants and 31% in preterm infants)

Conti...

4. Massive gut involvement that requires resection may result in short-bowel syndrome, with resultant failure to thrive or, worse still, pan-intestinal involvement that is incompatible with life.
5. Recurrence of the disease (9–14%)
6. Infants with stage II or III disease are commonly at risk of developing psychomotor retardation, although this may be more attributed to the high incidence of prematurity than a direct effect of NEC.

Table 35.2 Differential diagnosis of necrotizing enterocolitis

Feeding intolerance of prematurity

Spontaneous intestinal perforation

Malrotation/volvulus

Neonatal sepsis with ileus

Congenital intestinal obstruction (intestinal atresia,
Hirschsprung's disease)

Neonatal pseudomembranous colitis (rare)

Neonatal appendicitis (rare)

Diagnosis

- Investigations for diagnosis of NEC include the following.

- **Abdominal X-ray** may demonstrate:-
 - ✓ Dilated bowel loops
 - ✓ Edematous bowel wall
 - ✓ Ascites
 - ✓ Intramural gas (pneumatosis intestinalis)
 - ✓ Portal vein gas and free air in the peritoneal cavity (pneumoperitoneum).

Conti...

- **Signs of a pneumoperitoneum** include
 - ✓ **'Football sign'** (central collection of free air outlining the falciform ligament and umbilical arteries)
 - ✓ Localized gas in the right upper quadrant (the **'right upper quadrant gas sign'**), or
 - ✓ Gas on both sides of the bowel wall (**'Rigler's sign'**).
- The presence of a single/multiple loops of bowel retaining the same shape or position for 24–36h (**'fixed-loop sign'**) may suggest **full-thickness bowel necrosis**.

Conti...

➤ CBC

- ✓ In metabolic acidosis, the hemoglobin level may fall, owing to hemorrhage and sepsis, whilst leukocyte and platelet counts may be depressed, particularly in association with DIC.

➤ Bacteriological screening should include blood culture and collection of swabs from the nose, throat, umbilicus and rectum.

➤ Serum electrolyte

➤ Triads of NEC (Laboratory).....**Hyponatremia, Thrombocytopenia and Metabolic acidosis**

Treatment

Medical

- The initial management of NEC consists of stabilization of the infant and resting of the intestinal tract by means of discontinuation of enteral feeds for 10 days, total parenteral nutrition, decompression of the stomach by a nasogastric tube, fluid resuscitation, intravenous antibiotics for 7–14 days and correction of any metabolic derangement.

Surgery

- However, up to 50% of neonates with NEC develop advanced disease that requires **surgery**.

Prognosis

- The outcome from NEC is highly variable
 - ✓ With the **postoperative complication and mortality rates ranging between 10 and 70%.**

- These depend on:
 - (1) Stage of the disease
 - (2) Extent of disease involvement, with the lowest survival in pan-intestinal disease
 - (3) Birth weight
 - (4) Gestational age with 77% survival in term infants compared to 66% in preterm infants
 - (5) Presence of associated medical co-morbidities
 - (6) Modality of operative treatment.

- This affects the postoperative complication rate for multifocal NEC
 - ✓ (85% survival after resection and primary anastomosis

 - ✓ Compared to **50% with enterostomy alone**) but not isolated NEC.

Follow up of Preterm baby

- Follow weight gain at least 15-20gm/kg/day
- Follow for head circumference 1cm/wk
- Anthropometric measurement using corrected age
- Vision assessment
- Hearing assessment
- Brain ultrasound
- Development assessment
- Assess for anemia, rickets and bleeding tendencies

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